



OPEN Effect of graphene nanoplatelets on the behaviour of cementitious compounds: an experimental investigation

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Concrete has an inherent tendency towards brittle behaviour and micro-cracking, which can further degrade the durability in harsh environments. Graphene nanoplatelets (GNPs) have been recognized as highly potential nano-enhancements for improving the mechanical and durability performance of cement-based materials. While there is extensive literature evaluating the impact of GNPs on concrete, it is generally limited to low amounts of graphene, one or two properties assessed, and therefore, the overall impact of varying amounts of GNPs on the material remains unclear. The purpose of this research was to provide a thorough experimental assessment of how GNPs affect the fresh-state, mechanical and long-term durability properties of the concrete. Twenty-five different concretes at five strength levels (M20–M40) were made using GNPs at concentrations ranging from 0 to 20% by weight of cement. The experimental portion of this project included evaluations of fresh-state properties (workability and setting characteristics), mechanical properties (compressive strength, flexural strength, split tensile strength), and durability properties (water absorption, sulfate resistance, and rapid chloride ion permeation). The results show that incorporating GNP into cementitious composite systems significantly enhance their properties and performance. The compressive strength of the GNP containing systems reached around 47 Mpa while the flexural strength of the same systems reached 7.3 Mpa. The split tensile strength of these systems showed significant improvement in comparison to control systems. Additionally, the durability of the systems was significantly enhanced with respect to water absorption (reduced from around 4.5 to 1.6%), sulphate expansion (from 12.5% to 3.8%) and chloride permeability (reduced to approximately 650 coulombs) all demonstrating substantial pore refinement and resistance to aggressive ions. The above improvements have been attributed to the nano-filling effect of GNP; nucleation of hydrates during curing; and crack bridging effects caused by the presence of GNP. The collective effects of GNP provide microstructural density through the formation of a well bonded matrix and improved interface. The highest strength and durability were achieved with 20% GNP, however a GNP dosage range of 5–10% GNP was determined to be the optimal balance of performance and workability. These data demonstrate the strong potential of GNP as a means to develop high performance and durable nano engineered concrete systems for the development of advanced structural and infrastructures systems.

Keywords Graphene nanoplatelets, Cementitious composites, Mechanical enhancement, Durability performance, Nano-engineered concrete

Cement-based materials remain the most widely used general-purpose structural materials worldwide; however, their brittle nature, high porosity, and susceptibility to microcracking limit their long-term performance and structural reliability¹. Traditional concretes tend to have low crack resistance and therefore deteriorate

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rapidly under both mechanical and environmental loading². These shortcomings are further increased in the contemporary infrastructural systems that are vulnerable to hostile conditions such as sulphate-bearing soils, chlorides, freeze-thaw cycles, and varying mechanical forces. The recent research on nano-reinforced cementitious systems notes that the classical methods of enhancing strength, such as mineral admixtures and fibers are not enough to fill the micro-scales defects that cause early damage initiation^{3–6}. The lack of intrinsic sensing also prevents real-time damage diagnosis, leading to delayed maintenance and the failure of critical structures.

As the need for durable, self-sensing, and high-performance materials grows, considerable concern has been raised about graphene-based nanomaterials due to their outstanding tensile strength, electrical conductivity, and high specific surface area³. Recent studies indicate that graphene nanoplatelets and other types of carbon-based nanomaterials can contribute to significant enhancements to both the mechanical properties and durability as well as electrical conductivity of composite materials made from a variety of cementitious mixtures. The primary reasons for the enhanced mechanical behavior, durability and electrical conductivity of graphene-based nanocomposite materials is due to the “nano filling effect,” “crack bridging” mechanisms, and an increased rate of hydration. More recently, it has also been shown that graphene-based nanomaterials can significantly improve the mechanical and durability properties of composite cementitious systems by producing a denser microstructure and promoting the formation of hydrated cement paste. The use of graphene-based nanomaterials in cementitious composites has been investigated with some success; numerous reports indicate improvements in mechanical (compressive strength) and durability (resistance to cracking and permeability) characteristics of cement systems modified with graphene, which will ultimately lead to the development of advanced structural materials. These findings have been supported by many recent studies examining graphene-based reinforcement techniques for improving the mechanical and durability properties of cementitious composite systems.

Prior research shows that graphene nanoplatelets (GNPs) can optimise the geomorphology of C–S–H, improve hydration, reduce capillary porosity, and provide superior crack-bridging properties compared to traditional nanomaterials⁴. Many studies to date have focused on low-dosage graphene levels and have examined only a few properties (e.g., compressive strength or workability). There is very little data available on multi-grade concrete evaluations, particularly regarding wide dose ranges and durability performance indicators. The improvements in these areas are quantifiable: several studies show that if well-dispersed, compressive strength, flexural strength, and permeability may increase by 15–45% and 20–60%, respectively, and decrease by half. Additionally, the conduction network of graphene offers significant potential for developing smart cement composites that enable real-time strain and damage measurements, which are crucial for next-generation infrastructure⁵. Although the potential advantages of this performance are dependent upon both dosing and dispersion efficiency as well as interactions within the cement matrix⁶, a specific dosage range (0–20%) has been used to study the dose-dependent behaviour of graphene nanoplatelets in cement-based composite materials, so that the dosage-dependent behaviour of these nanoplatelets could be studied systematically. Other researchers have typically confined their dosage ranges for the examination of these materials to significantly lower values than those chosen in this study. This approach diminishes the likelihood of identifying any “threshold” or “saturation” effects at the higher doses.

Various studies based on past research have limited their scope to small amounts of GNP, typically 0.02–1% by mass of cement, thereby limiting the ability to observe dose-dependent behaviour at high nanoplatelet concentrations^{7–10}. Most of the research has focused on low graphene dosages (less than 1%) and on testing individual properties of the concrete, mainly compressive strength, or workability. This lack of research into the dose-dependent effects of graphene nanoplatelet dosages above 1% and their potential impacts on the fresh, mechanical, and durability performance of various concrete grades has limited our understanding of how these materials will perform across a wide range of applications. Additionally, the primary performance indicators of most of this prior research are primarily based on compressive strength or slump, while other critical performance indicators, including the durability of concrete, specifically water absorption, sulfate resistance, and chloride penetration, were completely ignored.

In previous research, dispersion methods were inconsistent, leading to agglomeration and ineffective integration with the matrix, and yielding contradictory findings regarding the actual impact of GNP on cement hydration and microstructural refinement². Moreover, fundamental functional parameters such as piezoresistive response, electrical stability, and strain sensing are far less studied, although the conductive properties of graphene are well known, and its potential as a structural health monitoring tool is also acknowledged⁸. The number of studies exploring the performance of various concrete grades is extremely low, and nearly none have endeavoured to construct a single composite model of the connection among graphene consumption, microstructural densification, mechanical behaviour, durability, and electromechanical sensing^{9–14}.

The purpose of this research is to close the knowledge gap described above through an experimental investigation of a wide range of properties, including fresh, mechanical, and durability properties, of multiple concrete grades with varying dosages of graphene nanoplatelets. In addition to information on the behaviour of GNPs as a function of dosage, this method will provide better information for performance optimization in graphene-modified cementitious composite systems. To address the basic limitations mentioned above, the present study formulated 25 graphene-reinforced concrete mixtures (grades) designated by mix IDs (M0-0 to M4-20) using a wide range of GNP dosages from 0% to 20%. A goal of the current study was to explore the effect of GNP dosages from 0% to 20% to determine whether GNPs exhibit dose-dependent behaviour, including a threshold or saturation point at which additional GNPs do not result in increased performance. High GNP dosages of up to 20% are not always economically viable for large-scale building projects; however, exploring this higher GNP dosage range will provide valuable insights into the performance limitations and optimization potential of GNP-modified cementitious composite systems. This expanded range will allow the researchers to determine whether there are threshold or saturation points at which the nanoparticles become too

crowded and begin to aggregate, as they would in low-dosage studies. The experimental program will measure the fresh properties, mechanical properties, durability performance, and microstructural characteristics of the specimens tested. Thus, it will cover several objectives. It demonstrated significant improvements, with compressive strength of 47 MPa, flexural strength of 7.3 MPa, a significantly lower water absorption of 1.6%, and a notable reduction in chloride-ion penetration under the RCP test. There was a steady improvement in sulphate resistance with increasing GNP content, suggesting greater chemical stability and a finer pore structure. Of note in this research is that both the piezoresistive properties and the strain-sensing capabilities are being examined, and a quantitative relationship among the amount of GNP used, the matrix conductivity arising from the formation of conductive networks, and the real-time electromechanical responses is being developed for the first time. In contrast to most other studies that have traditionally examined a restricted number of mixtures, this research will assess twenty-five combinations of mixtures composed of five distinct strength-grade concretes and five various GNP dosages. Thus, this broader experimental framework will provide a greater understanding of the dose-dependent behaviour, potential agglomeration thresholds, and performance optimization in GNP-modified cementitious composite systems. This research will also establish quantitative relationships among GNP dosage, pore refinement, improvements in mechanical properties, and durability performance. This will provide new insights into the macro-micro performance relationships in GNP reinforced cementitious materials. By integrating results from various performance indicators and dosage levels, this research will contribute to a better understanding of GNP's role in enhancing the structural performance and durability of cement-based materials. The collective gaps identified in previous studies are summarised in Fig. 1, highlighting the need for a comprehensive dosage-dependent investigation.

Materials

The materials that were used in this study included: ordinary portland cement, manufactured sand, crushed coarse aggregates and graphene nanoplatelets (GNP) as shown in Fig. 2. The representative image of constituent materials used in the experimental program of cement, fine aggregates, coarse aggregates, GNP and a superplasticizer. GNP was added to cement in percentages of 0, 2, 5, 10 and 20%. Research has documented that adding nanocarbon materials significantly improves the mechanical properties of concrete, the density of the matrix, the rate of hydration, and the ability of the material to resist crack formation^{3,4}. The aggregates were selected in accordance with the general grading requirements, as suggested by prior research on concrete technology for nano-reinforced systems^{5,6}. Mixes had a water-cement ratio of 0.36–0.50, similar to the literature, which has established that nano-additions elicit strong responses in low-to-medium w/c systems^{7,8}. A superplasticiser was added at the rate of 0.50–1.60% of the cement mass in the form of a polycarboxylate to offset the workability decrease caused by GNP, as was reported in nano-silica- and graphene-modified concretes⁹. In order to make up for the loss of workability that resulted from the increased surface area of graphene nanoplatelets, the amount of polycarboxylate-based superplasticizer added into the mix designs was varied. All materials studied in this research are listed in Table 1 along with their chemical and physical characteristics. The ability of the graphene nanoplatelets examined in this study to fill voids at the micro-level as well as bridge cracks was identical to the results of previous studies on other synthesized forms of graphene oxide (GON), GNP, carbon nanofibers, and CNT modified cementitious composites^{10,11}. Each of the graphene nanoplatelets used in this study consisted of

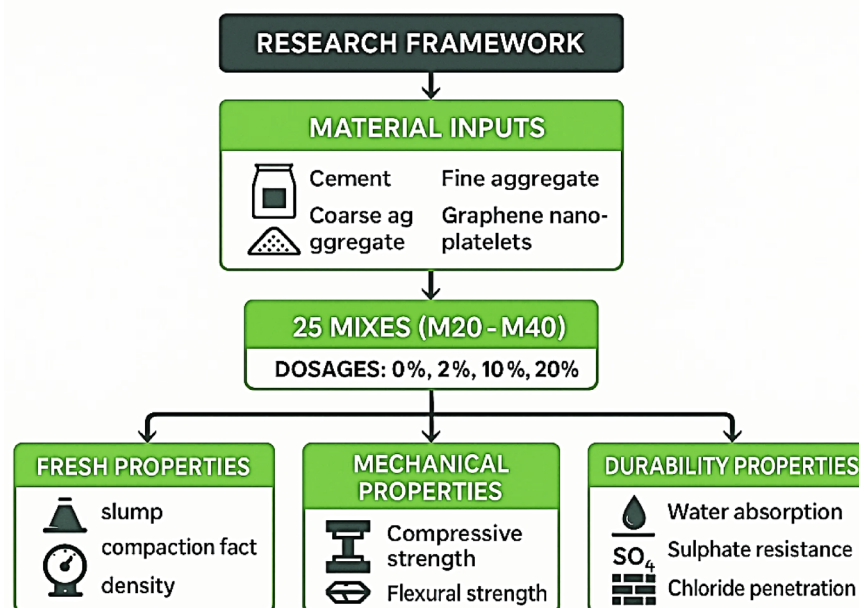


Fig. 1. Summary of past research trends on graphene nanoplatelet (GNP) applications in cementitious composites.

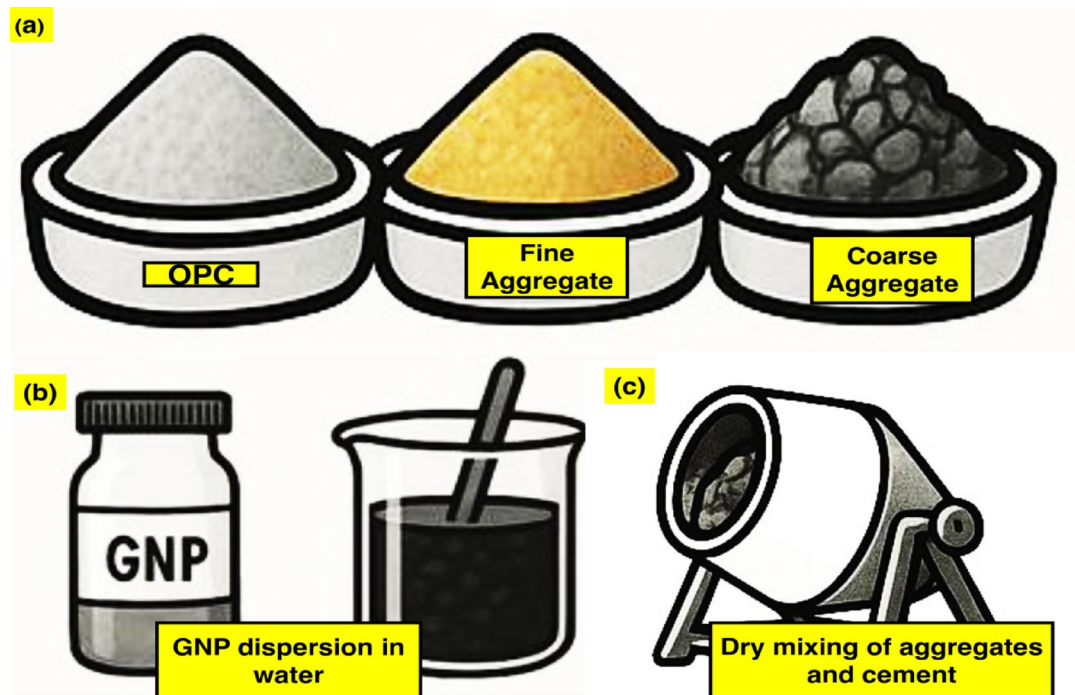


Fig. 2. Schematic illustration of (a) Constituent materials, (b) GNP dispersion in water, and (c) Dry mixing process in GNP-modified concrete.

multiple layers of graphene and had an average width of approximately 5 microns and a thickness of roughly 9 nanometers, resulting in a high-aspect-ratio geometry. The graphene nanoplatelets used in this study were also supplied as a powder with a purity greater than 98%, enabling maximum reinforcement potential and nanofilling behaviour within cement matrices.

The dispersion and deagglomeration were enhanced by mixing them in water with the addition of superplasticizer, a technique which is generally suggested in the existing literature to obtain homogeneous nano-reinforcement^{12,13}. Graphene nanoplatelets were dispersed in mixing water with polycarboxylate-based superplasticizers, then mechanically stirred using a high-shear mixer to avoid agglomerating them prior to being added to the mix. Although the authors did not use SEM or TEM (direct methods to characterize at a micro-level) to characterize their samples, it is well-documented that the high shear mixing and polycarboxylate-based superplasticizer combination will provide an enhanced level of dispersion of graphene nanoplatelets in cementitious systems. There are documented reports where the use of these combinations of materials has resulted in a significant reduction in agglomeration of graphene nanoplatelets and a uniform distribution of nanoscale reinforcement throughout the cement matrix. Therefore, to further improve the dispersion of graphene nanoplatelets and to prevent agglomeration, the graphene nanoplatelets were dispersed in mixing water that contained a polycarboxylate-based superplasticizer. The resulting suspension was then subjected to mechanical stirring using a high-shear laboratory mixer for approximately 10–15 min to ensure an even distribution of the nanoplatelets. The pre-dispersion process used in this study is commonly practised in the preparation of graphene-modified cementitious systems to achieve better nanoparticle distribution and minimize nanoparticle clustering¹⁷. A schematic illustration of the constituent materials, GNP dispersion process, and dry mixing procedure is shown in Fig. 2.

Methodology

The experimental project was based on an experimental plan of twenty-five concrete mixes, in five different strength levels (M20–M40), using five graphene nanoplatelet (GNP) content percentages of 0, 2, 5, 10, and 20% by weight of cement. Each mix design was prepared in accordance with the standard specifications of IS 10262:2019 and IS 456:2000. The test materials were first brought to saturated-surface-dry conditions before mixing. The GNPs were dispersed into the mixing water using a high shear mixer along with a polycarboxylate-based superplasticizer; this has been strongly recommended in earlier studies to prevent aggregation of nanoplatelets^{18,19}. Higher superplasticizer dosages were tested in the initial trial mixtures as well. However, it was decided that the final range would be chosen with an emphasis on ensuring sufficient workability to prevent segregation of ingredients while avoiding over-retarding the mixture. In terms of the dispersion of nanoparticles (GNPs), the first step involved mixing a suspension of GNPs and water containing the selected superplasticizer dosage. The suspension was then mixed for about 10–15 min in order to de-agglomerate the GNP particles before they were slowly added to the previously prepared mixture of dry cement and aggregate. Using this technique will improve the dispersion of GNPs and increase the likelihood that there is a more even distribution of GNP's within the entire concrete matrix. With five different GNP dosages tested in each of five different

Materials	Physical properties	Numerical value	Chemical properties (oxide/element %)	Numerical value
Ordinary Portland cement (OPC)	Specific gravity	3.12	CaO	63%
	Fineness	320 m ² /kg	SiO ₂	20.5%
	Bulk density	1440 kg/m ³	Al ₂ O ₃	6%
	Initial setting time	120 min	Fe ₂ O ₃	3%
	Final setting time	235 min	MgO	2%
	Avg. particle size	20 µm	SO ₃	2.5
	Colour	Grey	Alkalis (Na ₂ O + K ₂ O)	0.85
Fine aggregate (M-sand)	Specific gravity	2.62	SiO ₂	72.5%
	Water absorption	1.4%	Al ₂ O ₃	10%
	Bulk density	1750 kg/m ³	Fe ₂ O ₃	3.5%
	Fineness modulus	2.8	CaO	3%
	Moisture content	0.5%	MgO	1.5%
	Max size	4.75 mm	Other Traces	1.5%
Coarse aggregate (crushed granite)	Specific gravity	2.74	SiO ₂	65%
	Water absorption	0.7%	Al ₂ O ₃	16.5%
	Bulk density	1520 kg/m ³	Fe ₂ O ₃	4.5%
	Impact value	18%	CaO	2.5%
	Crushing value	23%	MgO	1.5%
	Max size	20 mm	Others	< 1%
Graphene nanoplatelets (GNP)	Specific gravity	2.0	Carbon (C)	95%
	Surface area	500 m ² /g	Oxygen (O)	2.5%
	Thickness	9 nm	Hydrogen (H)	0.75%
	Lateral size	5 µm	Sulphur (S)	< 0.5%
	Purity	> 98%	Metallic Traces	< 0.1%
	Average lateral platelet size	5 µm	Estimated aspect ratio ~ 550	–
	Electrical conductivity	10 ⁵ –10 ⁶ S/m	–	–
Mixing water	pH	7.25	Total Dissolved Solids	460 mg/L
	Hardness	270 mg/L	Chlorides	180 mg/L
	Density	1000 kg/m ³	Sulphates	180 mg/L
Superplasticizer (PCE-based)	Density	1.08 kg/L	Polymer Content	37.5
	pH	6	Sulphate Content	4%
	Solid content	37.5	Sodium (Na ⁺)	0.85%
	Dosage (of cement)	1.1%	Chloride (Cl ⁻)	0.08%

Table 1. Physical and chemical properties of raw materials used in the present study.

grades of concrete, the results provide a means to perform a sensitivity analysis of how dosage affects both fresh, mechanical and durability characteristics of concrete in addition to providing a means to evaluate trends and thresholds associated with performance. This GNP water interruption was gradually incorporated into the dry mix of cement, fine aggregate, and coarse aggregate.

A laboratory drum mixer was used to perform the mixing. Initially, dry aggregate materials (cement, fine aggregate, and coarse aggregate) were mixed for about 1–2 min. The prepared Graphene Nanoplatelet (GNP) Suspension was then slowly added to the mixer, which continued mixing for another 2–3 min until a homogeneous concrete mix was achieved. There were five major phases to this experiment: (i) dispersion of graphene nanoplatelets, (ii) preparation of concrete mixes, (iii) casting and curing of specimens, (iv) testing of the fresh properties of concrete, and (v) assessing the mechanical and durability performance of concrete. The detailed mix proportions for the twenty-five material combinations presented in this study are summarised in Table 2. Blending was performed for 3–4 min until uniformity was achieved. The fresh concrete was poured into steel moulds in two layers and then compacted by a vibrating table. To keep the appropriate workability of all of the different concrete mixtures, a gradual increase in Superplasticizer (SP) was added to each mixture as it contained increasing amounts of Graphene Nano-Platelets (GNP). A suitable SP dosage (approximately 0.50–1.60% of the mass of cement) was determined from initial trial mixes and the dosage ranges reported in other research studies on nanomaterial-modified concrete. Cubes were made to determine the compressive strength of the material, prisms were made to determine the flexural strength, and cylinders were made to determine the split tensile strength of the material. After casting, the specimens were demolded after 24 h and then cured in water at 27 ± 2 °C until tested at their respective ages.

Demoulding and water-curing of the specimens were performed after 24 h at 27 ± 2 °C^{1,8}. The general process involved in this experimental study was the evaluation of material characteristics, the creation of a graphene nanoplatelet (GNP) dispersion, the preparation of the concrete mixture, the formation of test specimens, the curing of the test specimens, and the evaluation of the fresh, mechanical, and durability properties of the test

Mix ID	Grade	Cement (kg/m ³)	Fine agg (kg/m ³)	Coarse agg (kg/m ³)	w/c	Water (kg/m ³)	GNP (% of cement mass)	GNP (g/m ³)	SP (% of cement)	SP (kg/m ³)
M0-0	M20	350	569	1050	0.50	175.0	0.00	0	0.50	1.75
M0-2	M20	350	569	1050	0.50	175.0	0.02	70	0.70	2.45
M0-5	M20	350	569	1050	0.50	175.0	0.05	175	1.00	3.50
M0-10	M20	350	569	1050	0.50	175.0	0.10	350	1.30	4.55
M0-20	M20	350	569	1050	0.50	175.0	0.20	700	1.60	5.60
M1-0	M25	400	650	1200	0.45	180.0	0.00	0	0.50	2.00
M1-2	M25	400	650	1200	0.45	180.0	0.02	80	0.70	2.80
M1-5	M25	400	650	1200	0.45	180.0	0.05	200	1.00	4.00
M1-10	M25	400	650	1200	0.45	180.0	0.10	400	1.30	5.20
M1-20	M25	400	650	1200	0.45	180.0	0.20	800	1.60	6.40
M2-0	M30	450	731	1350	0.40	180.0	0.00	0	0.50	2.25
M2-2	M30	450	731	1350	0.40	180.0	0.02	90	0.70	3.15
M2-5	M30	450	731	1350	0.40	180.0	0.05	225	1.00	4.50
M2-10	M30	450	731	1350	0.40	180.0	0.10	450	1.30	5.85
M2-20	M30	450	731	1350	0.40	180.0	0.20	900	1.60	7.20
M3-0	M35	500	813	1500	0.38	190.0	0.00	0	0.50	2.50
M3-2	M35	500	813	1500	0.38	190.0	0.02	100	0.70	3.50
M3-5	M35	500	813	1500	0.38	190.0	0.05	250	1.00	5.00
M3-10	M35	500	813	1500	0.38	190.0	0.10	500	1.30	6.50
M3-20	M35	500	813	1500	0.38	190.0	0.20	1000	1.60	8.00
M4-0	M40	550	894	1650	0.36	198.0	0.00	0	0.50	2.75
M4-2	M40	550	894	1650	0.36	198.0	0.02	110	0.70	3.85
M4-5	M40	550	894	1650	0.36	198.0	0.05	275	1.00	5.50
M4-10	M40	550	894	1650	0.36	198.0	0.10	550	1.30	7.15
M4-20	M40	550	894	1650	0.36	198.0	0.20	1100	1.60	8.80

Table 2. Mix Proportions for All Concrete Mix IDs (M0-0 to M4-20) Incorporating 0–20% GNP. The superplasticizer dosage was adjusted to compensate for the reduction in workability caused by increasing graphene nanoplatelet content.

specimens. The initial step included the dispersion of graphene nanoplatelets into a solution of water that contained the desired amount of polycarboxylate-based superplasticizer. This suspension was then mechanically stirred by utilizing a high shear mixer to ensure an equal distribution of the GNP in the solution and to reduce the potential for the GNP to agglomerate prior to introduction into the concrete mixture.

Compaction factor tests and slump tests, as per IS 1199:2018, were conducted to determine the workability of the fresh material. The slump settlement was calculated as

$$S = h_1 - h_2$$

where h_1 represents the initial cone height and h_2 represents the height after settlement. The compaction factor was calculated as

$$CF = \frac{W_1}{W_2}$$

where W_1 is the partially compacted concrete weight, and W_2 is the fully compacted concrete weight. The fresh density was established by calculating the proportion of mass to volume,

$$\rho = \frac{M}{V},$$

where M is the mass of the concrete sample, and V is the corresponding volume. The rising trend in density with increasing GNP dose verified the micro-filling effects reported in earlier works.

Setting time was determined using the Vicat apparatus as per IS 4031 (Part 5): 2019⁵⁶. For the setting time test, cement paste mixtures were prepared with cement, water, and graphene nanoplatelets at the same dosage levels as in the concrete mixes. A small amount of polycarboxylate-based superplasticizer was incorporated where necessary to maintain workable paste consistency when GNPs were added. Mechanical strength characteristics were tested according to the standard procedures outlined in IS 516:1959⁵⁷ for compressive and flexural strength, and in IS 5816:1999⁵⁸ for split tensile strength. Durability tests performed were water absorption, rapid chloride penetration, and sulphate resistance. Absorption of water measured by using the ASTM C642-21, in which the percentage absorption was calculated as.

$$WA = \frac{W_s - W_d}{W_d} \times 100,$$

where W_s and W_d were the saturated and oven-dry weight, respectively⁵⁹. Sulphate attack resistance was tested by using ASTM C1012-18 by immersing the sample in a 5% Na_2SO_4 solution and by monitoring expansion as

$$\text{Exp} = \frac{L_t - L_0}{L_0} \times 100^{60}$$

Chloride permeability was determined in accordance with ASTM C1202-19 by applying a 60 V potential for 6 h, as reported in previous studies⁶¹. Each test was conducted in accordance with the procedures outlined in the respective standards to ensure the consistency and reliability of the experimental results. The complete mix proportions for all 25 concrete mixtures are presented in Fig. 3.

Results and discussions

The experimental results are presented in three main categories: fresh properties, mechanical performance, and durability characteristics. This structure facilitates a clear interpretation of how graphene nanoplatelet dosage influences different aspects of concrete behaviour.

Slump test

The slump test results for all mix IDs show a progressive decrease in workability with increasing graphene nanoplatelet (GNP) content, regardless of concrete grade. For control mixes (0%), slump values ranged from 78 mm for the lower grades to 61 mm for the higher grades. The relationship between GNP dosage and workability trend in concrete is shown in Fig. 4 for various grades of mix. For mixes produced in accordance with M20-M40 strength level, it can be seen that the decrease in slump as the GNP dosage increases will follow a similar trend regardless of the strength grade of the concrete. However, mixes with high cement content showed somewhat greater improvement in workability due to the increased amount of paste in the mixture. Slump values decreased



Fig. 3. Mix Design Details for All Concrete Mix IDs (M0-0 to M4-20) Incorporating 0–20% GNP.

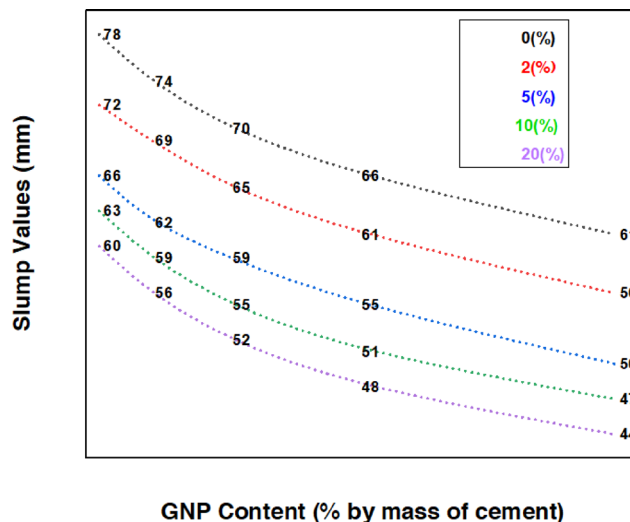


Fig. 4. Variation of Slump Values with Different GNP Dosages Across All Mix Grades (M0–M4).

from 72 mm to 56 mm when using a 2% GNP by mass of cement to 66 mm–50 mm when a 5% GNP by mass of cement was used. The behavior of mixes at the 2% GNP dosage are slightly different than those at the 5% GNP dosage; this may be attributed to lower concentrations of graphene nanoplatelet, where the primary influence of the particles on hydration kinetics is realized prior to significant reinforcement effects being developed in the concrete matrix. Mixes with 10 GNP by mass of cement showed slump values ranging from 63 to 47 mm, and the highest dosage of 20 resulted in the lowest slump values of 60 to 44 mm. The continuous downward trend confirms that the occurrence of GNP has a significant effect on the fresh-state behaviour, with the main effects being an increase in surface area and reduced flowability. These results are consistent with the proportional reduction in all the mix series (M0–M4) in the slump graph plotted.

The slump reduction with GNP content is progressive and aligns with trends reported in nano-reinforced cementitious materials. The decrease in workability can be linked to the extremely high specific surface area of nanoplatelets, which results in a rise in water demand, rapid absorption of free water, and reduced flowability²⁰. The amount of Polycarboxylate-Based Superplasticizer (SP) added to the mix design was increased as the addition of graphene nanoplatelets reduced the workability of the mixture. The increase in SP dosage provided an improvement in the workability of the mixture. However, additional increases in the SP dosage beyond that selected, had minimal effect on restoring the slump value of the mixture; therefore, it may have also affected the overall stability of the mixture. The SP dosage was therefore limited to a range that would provide a stable workable mixture. Similar slump reductions have been reported when using nanomaterials such as nano silica, graphene oxide, and carbon-based nanofillers, which tend to agglomerate in the cement matrix and increase internal friction²¹. Studies have also shown that graphene-enhanced mixes need a higher dosage of superplasticizer to retain desirable rheological behaviour⁹, which is consistent with SP modifications as shown in the present mix design table. The slump reduction observed from 0% to 20% GNP by mass of cement in the present study is similar to the workability losses observed in nano-modified UHPC and HPC systems, where particle interlocking, surface energy effects, and decreased paste lubrication are the predominant factors. The variability observed in the experimental measurements is represented by error bars in the corresponding figures, indicating the spread of replicate test results. Moreover, the extent of slump loss is greater at higher grades (M3–M4), which is in line with the literature, which states that mixes with lower w/c ratios exhibit greater sensitivity to the addition of nanomaterials^{22,23}. The significant reduction in slump values as GNP dosage increases is shown in Fig. 4.

Compaction factor test

The compaction factor test results show a continuous decrease in workability as GNP dosage increases across all mix IDs. Control Mixes 0% GNP by mass of cement had the highest compaction factors, which ranged from 0.92 in lower grades of concretes to about 0.84 in higher grades of concretes. The results presented in Fig. 5 confirm the consistent reduction in compaction factor with increasing GNP dosage. Upon the incorporation of 2% GNP by mass of cement, the compaction factor dropped to 0.90–0.82, while 5% GNP by mass of cement mixes yielded results from 0.88 to 0.80. At 10% GNP by mass of cement, the compaction factor further decreased to about 0.87–0.79, and the maximum dosage of 20% resulted in the lowest values, ranging from 0.84 to 0.78. This homogeneous decreasing trend is similar to that observed in the compaction factor test and confirms the conclusion that the addition of nanoplatelets reduces the flowability of fresh concrete.

The continuous decrease in the compaction factor with increasing GNP content is consistent with previous research on nano-engineered cementitious systems, in which reduced workability was attributed to the high surface area, hydrophilic nature, and tendency to agglomerate of nanomaterials²⁴. The addition of GNPs increases internal friction and reduces the amount of free water available for lubrication, resulting in a measurable decrease in compaction efficiency, analogous to observations in mixes containing graphene oxide, nano-silica,

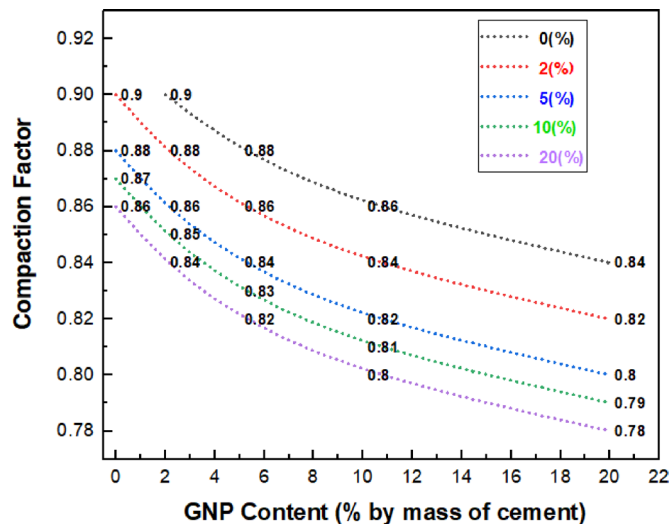


Fig. 5. Effect of GNP content on compaction factor for all mix IDs.

and carbon nanotubes^{25,26}. Further investigations have reported that platelet-shaped graphene particles disrupt particle packing, thereby increasing resistance to compaction, especially at dosages greater than 5%^{27,28}. This behaviour is enhanced in concretes with low water-to-cement ratios, consistent with the lower compaction factors of the higher-grade mixes (M3 and M4) in the present study²⁹. Similar behaviours of nano-inclusions in the microstructure of UHPC and HPC systems were previously shown to increase microstructural densification at the expense of fresh-state behaviour³⁰. Overall, the observed trend is consistent with established research and confirms that increasing GNP dosage significantly reduces the compaction capability of cementitious composites. This decrease in compaction factor with increasing GNP content is clear in Fig. 5.

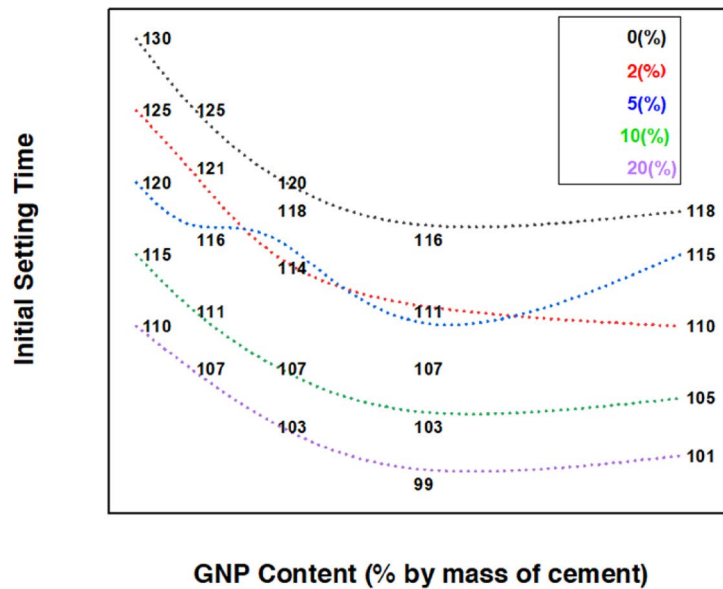
Setting time test

The initial and final setting times of the mixes showed a continuous decrease with increasing GNP content across all ID series. For the control mixes with 0% cement, the initial setting time ranged from 118 to 130 minutes, while the final setting time ranged from 248 to 260 minutes. The addition of 2% GNP by mass of cement decreased the initial setting time to 110–125 min and the final setting time to 232–250 min. At a 5% GNP by mass of cement, these values were further reduced to approximately 120–115 minutes for the initial setting time and 240–239 minutes for the final setting time. Mixtures with a GNP concentration of 10% exhibited a greater reduction, with the initial setting time decreasing from 115 to 105 minutes and the final setting time from 230 to 220 minutes. Graphene nanoplates (GNPs)' increased surface area enables them to provide additional nucleation sites for hydration products such as C-S-H gel. This allows hydration products to form faster than they would without GNPs, resulting in the formation of C-S-H gel and a hardened cement paste at an earlier age. Similar effects were observed in previous studies using graphene-based nanomaterials and nano-silica in cement-based materials. In those studies, the addition of nanoparticles enhanced the rate of hydration and enabled faster microstructural formation in the early stages of curing. The lowest values were measured at 20% GNP by mass of cement, with initial setting times ranging from 110 to 101 min, and final setting times ranging from 220 to 211 min. Overall, the trend confirms the accelerating effect of GNP on the setting process, with the most significant effect being observed in higher-grade mixtures.

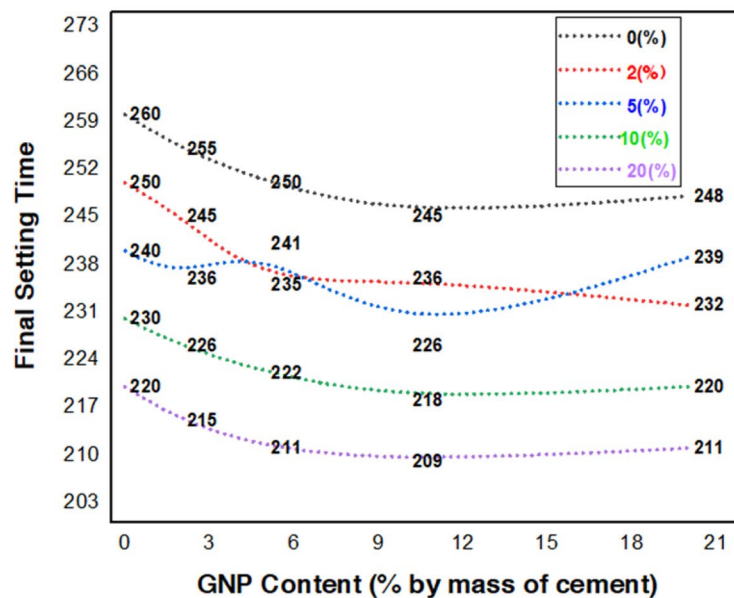
The decrease in the initial and final setting times with increasing GNP dosage is consistent with other studies on graphene-based nanomaterials in cementitious systems. This acceleration is typically attributed to the nucleation activity of GNP, thereby providing additional sites for the rapid formation of C-S-H gel³¹. Additional investigations have shown that graphene derivatives enhance hydration kinetics by increasing the dispersion of binder particles and promoting early-age microstructural refinement³². The decrease in initial and final setting times is associated with the behaviour reported for concretes modified with graphene oxide and nano-silica, in which nano-inclusions increase the reaction rate due to their higher surface energy and catalytic effects³³. The observed increase in setting time at later mix IDs for specimens containing 0% and 5% GNP by mass of cement can be explained by variations in water-to-cement ratio, consistent with previous findings on w/c sensitivity in hydration development³⁴. Moreover, studies on UHPC and HPC systems have shown that the use of nanomaterials enables the structuring and early stiffening of the paste to occur more quickly, which supports the performance observed in the current mix series³⁵. The accelerating outcome of GNP on the initial and final setting times is demonstrated in Fig. 6.

Fresh density test

The fresh densities of all mixes increased significantly upon addition of GNP, with the maximum values observed at 20% GNP by mass of cement. For the control mixes (0%), the fresh density ranged between 2270 and 2290 kg/m³, while mixes with 2% GNP by mass of cement exhibited an increase to approximately 2300–2360 kg/m³. The 5% GNP by mass of cement mixes continued this trend, reaching densities of 2330–2350 kg/m³. Mixes



(a)



(b)

Fig. 6. Initial and final setting time variation with increasing GNP dosage.

with 10% GNP by mass of cement yielded higher results, ranging from 2360 to 2400 kg per cubic meter. The highest densities were recorded in the GNP mixes with 20% content, which ranged from 2400 to 2430 kg/m³. Graphene's ability to fill nanoscale voids in cement paste and increase its density is responsible for the increase in fresh density. In addition to increasing the paste density, the addition of GNP may also reduce the amount of trapped air in the mixture. Future studies may measure the air content of fresh mixtures to better understand how GNP affects air-void characteristics. Future studies may include measuring the air content of fresh mixtures as part of an investigation into the effects of GNP on fresh density, air-void characteristics, and overall variation in density in the fresh state. The results of this study indicate that while it is possible to produce high-strength concrete with a low water-to-cement ratio (w/c), the use of superplasticizers may be required in order to obtain a workable mixture. This is consistent with the fact that the flowability of a mixture decreases as the w/c ratio

decreases and as the cement volume fraction increases. Overall, the results show that as GNP content increases, particle packing increases, which accounts for the higher fresh density across all mix identifiers.

The rise in density with the fresh density to higher GNP dosage is in line with the available literature on nano-modified cementitious composites. This performance can be explained by the fact that GNPs are capable of filling micro-voids, optimizing the density of the solid matrix, and lowering the level of entrapped air^{36,37}. It has been demonstrated that incorporating graphene derivatives enhances matrix compaction due to their high specific surface area and layered structure, allowing a denser arrangement of structures⁸. Studies of graphene oxide and nano silica have also shown that nanomaterials act as micro fillers, decreasing porosity and increasing the total density of fresh concrete³⁸. The peaks of higher densities at 10–20% GNP by mass of cement are consistent with results indicating that, as nanoparticle content increases, packing efficiency rises until a threshold dose is reached³⁹. The patterns of various mix IDs in this work demonstrate the behaviour of nano-enhanced HPC and UHPC systems, with reductions in void content and increases in fresh density observed. The variation in fresh density with increasing GNP dosage is graphically depicted in Fig. 7.

Compressive strength test

All mix series showed increases in compressive strength compared to controls, with the greatest increases observed at the highest nanoplatelet contents. The lowest strengths (~20–22 MPa) were observed in the control mixes; the 2% and 5% GNP by mass of cement mixes showed moderate improvements (peaks ~ at 32–33 MPa, respectively). As seen in this transitional behaviour at a 2% GNP dosage, it seems the particle density may still be insufficient to trigger the high effectiveness of mechanisms (i.e., crack bridging, pore refinement, matrix densification) at higher nanoparticle densities. Trends related to compressive strength development for each GNP content level are depicted in Fig. 8. The impact of varying GNP percentages upon the flexural strength of each respective mix is demonstrated in Fig. 9. In comparing the different mixes, while the beneficial effect of adding graphene nanoplatelets to compressive strength has been consistently demonstrated for each respective mix, the degree of increase in strength was somewhat dependent upon both the overall chemical make-up of the respective mix as well as the W/C ratios of each mix grade.

The values of the 10% GNP by mass of cement mixes were higher (~35–41 MPa), and the largest compressive strengths were recorded in the 20% GNP by mass of cement mixes (peaking near 47 MPa and ending around 44 MPa). A prominent mid-series peak is visible at certain dosages (e.g., 2% and 20% around mix IDs ~9–12), with minor decreases at later mix IDs. These tendencies align with the formulations and admixture modifications presented in the mix design table.

This improvement in strength with the increase in GNP content is in line with past indexed literature that characterizes nanomaterials as micro-fillers and nucleation sites, which enhance the acceleration of hydration and refinement of the C–S–H construction⁴⁰. The mixes studied here are for differing strength grades as per the IS 10262:2019 guidelines. These strength grades are differentiated by varying amounts of cement and water-cement ratios. While GNP's will certainly enhance the mechanical property improvements from these mixes, it is also possible that other aspects of the mix (i.e., cement content, water-cement ratio, superplasticizer amount) could impact the actual strength values of the mixes. However, the fact that similar trends occur within each strength grade indicates that GNP does provide a significant contribution to the enhanced mechanical performance of the mixes.

The enhancement in compressive strength observed in the present study is supported not only by general mechanisms reported in the literature but also by direct experimental evidence obtained in this study. The

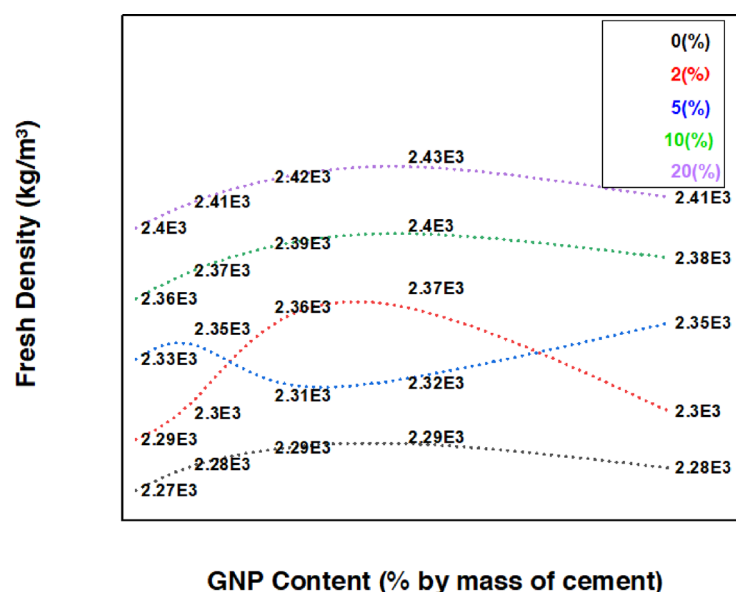


Fig. 7. Fresh density of concrete mixes as influenced by GNP percentage.

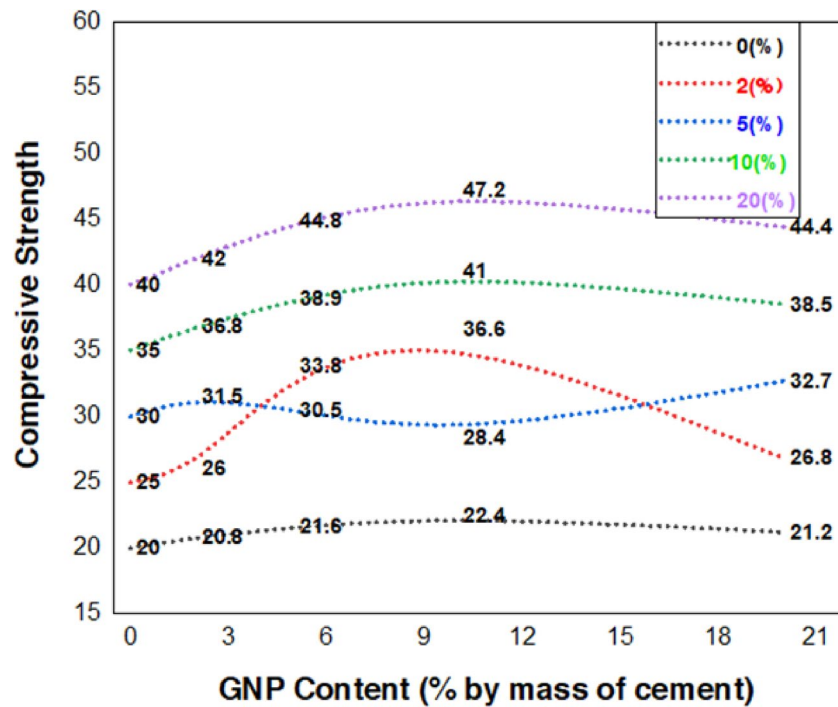


Fig. 8. Compressive strength development for different mix IDs with 0–20% GNP.

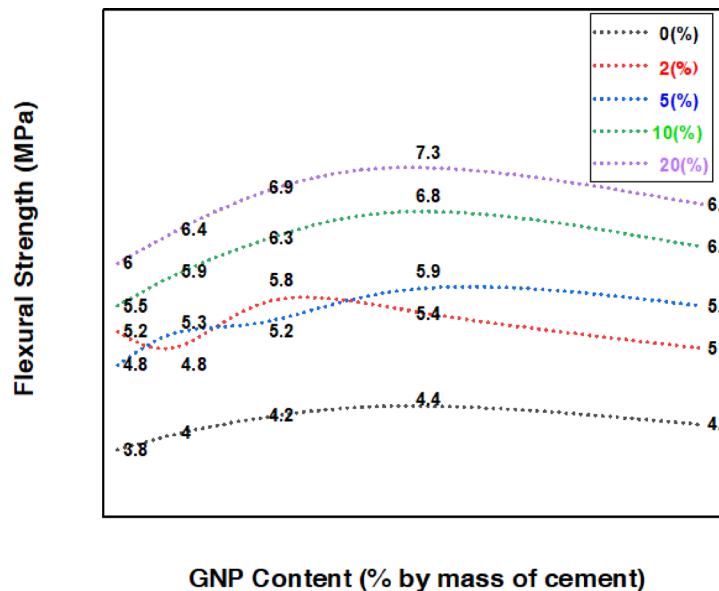


Fig. 9. Flexural strength variation of concrete mixes with increasing GNP dosage.

SEM analysis (Microstructural observation (SEM–EDS and XRD analysis) section) reveals a denser and more compact microstructure in GNP-modified specimens, indicating reduced pore connectivity and improved particle packing. This observation substantiates the nano-filling effect⁴¹, where graphene nanoplatelets occupy microvoids and refine the pore structure. Furthermore, EDS and XRD results confirm the presence and enhanced formation of C–S–H phases, suggesting that GNPs act as nucleation sites, accelerating hydration reactions.

The progressive increase in compressive strength with GNP dosage is therefore attributed to a combination of experimentally supported factors: (i) microstructural densification evidenced by SEM imaging, (ii) enhanced hydration confirmed through phase identification, and (iii) reduced porosity inferred from durability indicators such as decreased water absorption and chloride permeability. While crack-bridging effects are widely reported in the literature, their direct verification at the nanoscale was not performed in this study; however, the improved post-peak behaviour and strength gains indirectly support their contribution.

The presence of graphene nanoplatelets increases the number of nucleation sites for C–S–H gel formation, accelerating hydration and resulting in a denser microstructure that transfers loads more effectively. A dosage-dependent effect is also observed in several studies: small additions enhance strength by increasing the matrix density, but large additions can yield either further improvement or no improvement, depending on how the material is dispersed⁴². The mid-series peaks and the following minor strength reduction in certain mix IDs are probably the consequence of opposing positive nucleation/packing influences and negative dispersion agglomeration or ineffective use of superplasticizers at later mixes. The decreasing w/c mixes appear to be more affected by GNP content, as enhanced strengths become more evident in dense matrices, which aligns with the finding that low w/c systems are better suited to respond to nanofillers. Lastly, the relevant mix adjustments (cement, water, and SP in the mix design) affect the observed behaviour. Higher doses of superplasticiser were added to the mix table to mobilise the paste and enable the strength benefits of GNPs to be observed, as reported in previous research. Similar enhancements in compressive strength have also been reported in studies on cement-based composites with nano-graphite platelets, as nano-scale carbon reinforcement significantly increased the hydration rate and reduced the number of micro-cracks in the cement matrix. This study is consistent with other research showing that nanoscale carbon-based reinforcements increase matrix density and enhance load transfer. The compressive strength enhancement observed for higher GNP dosages is shown in Fig. 8.

Flexural strength test

The four mixes demonstrated improved flexural strength of the samples, as reflected in the GNP calculation (with increasing dosage, the degree of enhancement was distinct). The control mix (0%) had the lowest strengths, ranging from approximately 3.8 to 4.4 MPa between the mix IDs. The accumulation of 2% GNP by mass of cement upgraded the flexural strength to about 5.2 to 5.9 MPa, while the addition of 5% GNP by mass of cement gave slightly higher values ranging from 4.8 to 5.9 MPa. The 10% GNP by mass of cement mixes showed significant improvements, reaching 5.5–6.8 MPa. The best flexural strengths were achieved at 20% GNP by mass of cement, with values around 7 MPa at their peak and finishing close to 6.7 MPa. These results indicate that GNP addition positively enhances flexural performance by enhancing crack bridging and matrix reinforcement.

The observed increase in flexural strength with increasing GNP content is consistent with earlier reports on nano-reinforced cement composites with graphene-based materials, which enhance load transfer and the efficiency of crack bridging⁴³. The layered morphology and high tensile modulus of GNPs enable the arrest of microcracks and increase the ductility of the cement matrix, which explains the considerable increase in flexural strength observed at 10–20% GNP by mass of cement inclusion⁴⁴.

The improvement in flexural strength is consistent with both the experimental observations and the microstructural evidence obtained in this study. SEM images (Microstructural observation (SEM–EDS and XRD analysis) section) indicate a more homogeneous and compact matrix in GNP-modified concrete, thereby enhancing stress transfer within the matrix. This densified microstructure contributes to improved resistance against crack initiation and propagation under flexural loading.

Although direct visualization of crack-bridging at the nanoscale was not conducted, the significant increase in flexural strength, particularly at higher GNP dosages, suggests the presence of effective reinforcement mechanisms. Additionally, the reduction in pore connectivity, as evidenced by lower water absorption and chloride permeability, indicates improved matrix integrity, which further enhances flexural performance. Therefore, the flexural strength enhancement is interpreted as a combined effect of matrix densification, supported by SEM observations, and improved interfacial bonding, inferred from durability performance.

Similar improvements have been reported for graphene oxide and carbon nanofibers since these nanomaterials improve fiber-matrix adhesion and interfacial toughness⁴⁵. The highest flexural strength value at the middle of the mix IDs (e.g., 7–12) indicates optimal dispersion and efficient stress transfer, a trend also observed in other studies on the nano-modification of UHPC and HPC⁴⁶. The mechanism that bridges cracks improves the fracture energy of the cement matrix by delaying crack propagation. As a result, the mechanical properties of the cement matrix are enhanced in both flexural performance and toughness. Improvements in flexural performance have similarly been reported for fiber reinforced cementitious composites; these improvements were the result of reinforcement mechanisms which included both crack bridging and stress redistribution and resulted in improved resistance to failure resulting from bending loads. At higher mix IDs, the slight reduction in flexural strength can be attributed to the paste's limited flowability or to partial agglomeration of the nanoplatelets, as reported in studies of dispersion sensitivity in nanocomposites. The trend of flexural strength improvement with increasing GNP dosage is also consistent across the different concrete grades, indicating that the crack-bridging and reinforcement effects of graphene nanoplatelets are effective across a wide range of matrix strengths. The significant increase in flexural strength with 10–20% GNP by mass of cement is obtainable in Fig. 9.

Split tensile strength

The split tensile strength increased with the incorporation of GNP across all mix series, and the maximum values were consistently observed at the highest nanoplatelet content. Control mixes with 0% GNP by mass of cement had the lowest tensile strengths (~2.5–2.9 MPa), while mixes with 2% and 5% GNP by mass of cement showed moderate improvements (peaks ~3.3–3.2 MPa). As the graphene nanoplatelet dosage increases beyond 2%, the reinforcement mechanisms associated with nanoparticle incorporation become more dominant, leading to more consistent improvements in tensile performance. Mixes with 10% GNP by mass of cement achieved higher strengths (~3.4–4.0 MPa), while mixes with 20% GNP by mass of cement achieved the highest strengths in the series, reaching a maximum of almost 7.3 MPa at mid-series mix IDs and tapering off to around 6.7 MPa. A mid-series peak (mix IDs ~9–12) is clear for most dosages, followed by a slight decline at later mix IDs, indicating the sum of dosage, dispersion, and change (mix) effects. Overall, the data suggest a clear, dose-dependent improvement in split-tensile performance with increasing GNP content.

The increase in split tensile strength with GNP incorporation is supported by both macroscopic test results and microstructural observations. The SEM analysis (Microstructural observation (SEM–EDS and XRD analysis) section) demonstrates a refined, dense microstructure with reduced voids, thereby enhancing load transfer across the cement matrix. This improved matrix continuity contributes directly to higher tensile resistance. Furthermore, the significant reduction in permeability-related parameters, such as water absorption and chloride-ion penetration, indicates a less porous, more cohesive internal structure. This densification of the interfacial transition zone (ITZ) is likely to improve bonding between aggregate and paste, thereby enhancing tensile performance⁴⁷. While crack-bridging is a commonly cited mechanism, its presence in this study is inferred indirectly through strength improvements rather than directly observed⁴⁴. Hence, the interpretation is grounded primarily in experimentally verified microstructural densification and improved matrix integrity.

Numerous authors report a dosage-dependent response, with low- to moderate-GNP additions improving matrix toughness and bonding, while very high dosages provide additional benefits if dispersion is adequately maintained; otherwise, agglomeration can reduce effectiveness in this regard⁴⁸. The mid-series peak and slight declines for mix identifiers as we go later in the series are likely to indicate an optimum dispersion window observed within a mix composition, beyond which decreased workability or insufficient compensation from the superplasticizer cannot result in additional improvements⁴⁹. Sensitivity to water/cement and superplasticiser contents is also often reported; low water/cement mixes tend to exhibit more pronounced tensile enhancements due to the nanofillers, as the already dense matrix allows for a greater relative effect of crack bridging. Additionally, micro-filling of GNPs reduces porosity and narrows the interfacial transition zone, thereby enhancing tensile strength and toughness under splitting loads. The enhancement in the interfacial transition zone (ITZ) may be due to GNPs improving the density of hydrate products around aggregate surfaces; further, GNPs can fill voids in the microstructure around aggregate surfaces, thereby increasing the overall density of the ITZ. As the interfacial transition zone is generally the weakest region in conventional concrete, increasing its density improves stress transfer from the aggregate to the surrounding hydrated cement paste. Therefore, the enhancement of tensile strength is directly related to the increased density of the interfacial transition zone and the reduction of local stresses at the aggregate interface. Research on nano-reinforced cementitious composites demonstrates that carbon-based nanomaterials can enhance tensile strength by improving interfacial bonding in the interfacial transition zone and reducing microcrack propagation. The dosage-dependent improvement in split tensile strength is illustrated in Fig. 10.

Across the various concrete grades, the addition of graphene nanoplatelets consistently improves tensile strength, although the relative improvement is slightly more pronounced in mixes with lower initial matrix strength.

Water absorption test

The water absorption values of all mixes decreased consistently with increasing GNP content, indicating increased impermeability and reduced porosity. The durability performance indicators, including water absorption, sulphate expansion, and chloride penetration resistance, are presented in Figs. 11, 12 and 13. The Control mixes with 0% had the highest absorption values, ranging from about 3.4% to 4.5%. In contrast, 2% GNP by mass of cement mixes narrowed the range to approximately 4.0–3.0%. The relatively modest improvement observed at the 2% dosage may therefore represent an initial nucleation effect, while higher dosages contribute more significantly to pore refinement and microstructural densification. Further declines were noted for 5% GNP by

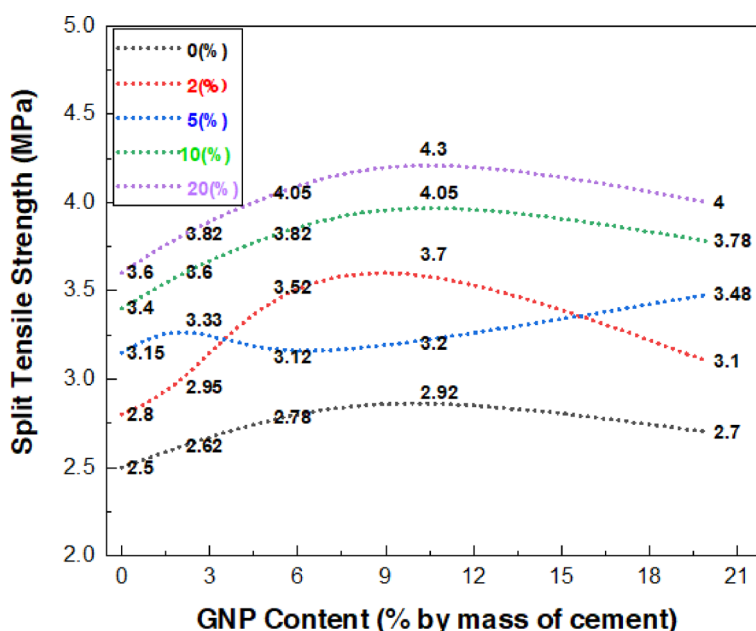


Fig. 10. Split tensile strength trends for mix IDs containing 0–20% GNP.

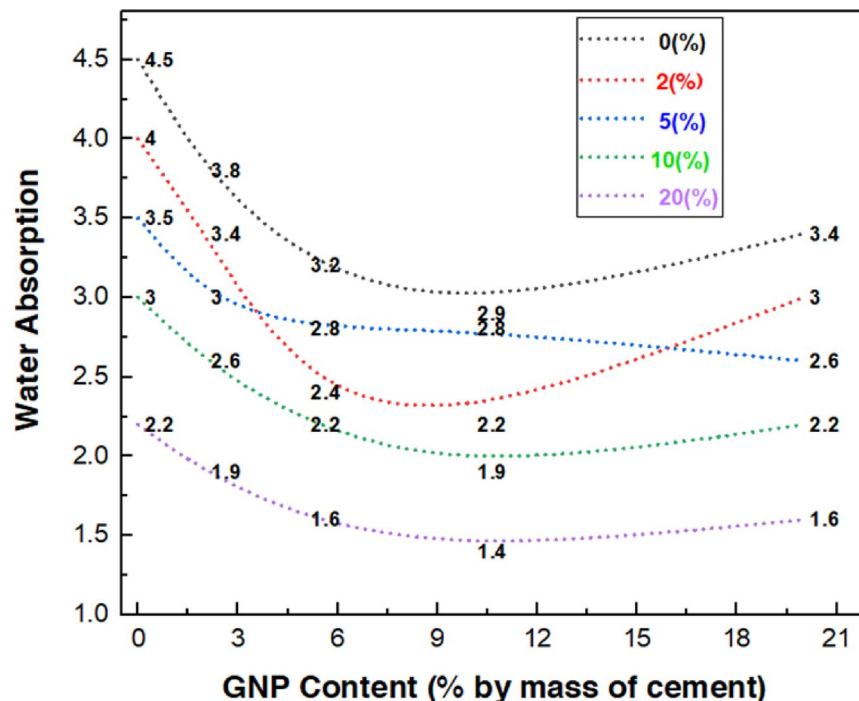


Fig. 11. Water Absorption Performance of GNP-Modified Concrete.

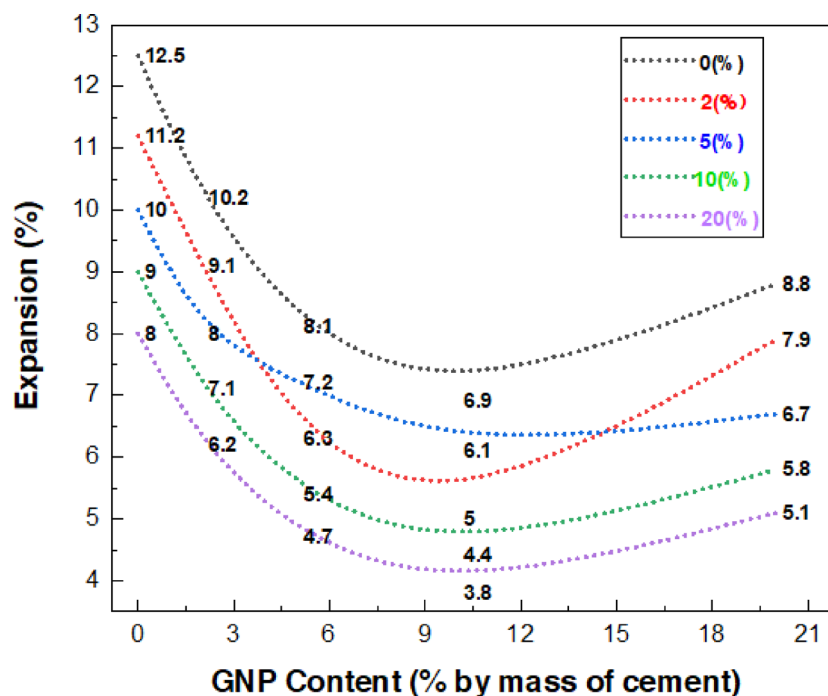


Fig. 12. Sulphate attack resistance (Expansion %) for all mix IDs at varying GNP contents.

mass of cement mixes (3.5–2.6%) and 10% GNP by mass of cement mixes (3.0–2.2%). The lowest absorption values were observed in the 20% GNP-by-mass cement mixes, ranging from 2.2% (early mix IDs) to nearly 1.6% (later mix IDs). These trends clearly indicate that, across all mix ID series, higher GNP dosage leads to greater densification and moisture resistance of the composite.

The observed decrease in water absorption with more GNP content is in good agreement with data obtained from previous research on graphene-based and nano-modified cementitious materials. Graphene nanoplatelets enhance microstructural packing, reduce pore connectivity, and fill micro-voids, thereby reducing permeability

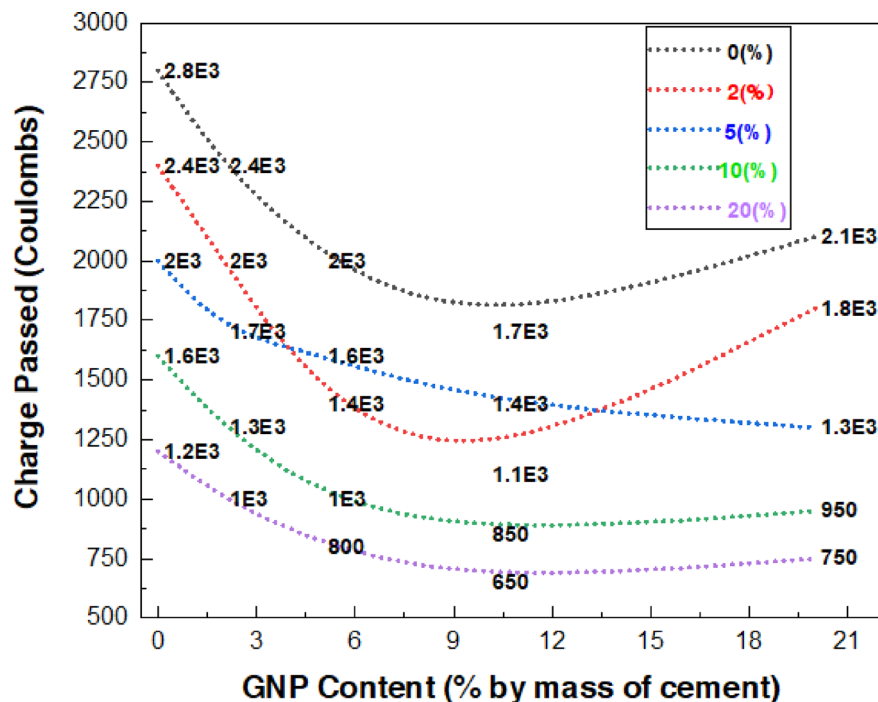


Fig. 13. Rapid Chloride Penetration Test (RCPT) Results Showing Charge Passed (Coulombs) vs. GNP Dosage.

and moisture penetration⁵⁰. The reduction in pore connectivity also increases the tortuosity of transport pathways for water and aggressive ions. As a result, the ingress of moisture, sulphates, and chlorides becomes significantly restricted, which explains the improved durability indicators observed in the present study. The observed significant decrease at 10–20% GNP is consistent with studies showing that graphene derivatives improve the pore structure and tortuosity of the water pathway⁵¹. Comparable improvements in the impermeability have been reported in concretes modified with graphene oxide, nano-silica and carbon nanotubes, where nanofillers reduce both capillary pores and the gel pore volume⁵². From an engineering perspective, reduced water absorption indicates improved resistance to moisture ingress, which can significantly delay deterioration processes such as reinforcement corrosion and chemical attack in aggressive environments. The decrease in absorption in the mixes at the early stage is also consistent with reports that GNP is identified as accelerating hydration and densifying the C-S-H gel, thereby reducing porosity within the first 28 days. Moreover, the lower water absorption observed at higher GNP dosages corresponds to increased barrier capacity and greater impermeability, which are typically associated with nano-reinforced HPC/UHPC systems. The continuous reduction in water absorption with higher GNP dosage is shown in Fig. 11.

Sulphate attack resistance

The consequences of sulphate attack show a clear decrease in deterioration as GNP content increases across all mix identifiers. The control mix (0%) showed the greatest expansion and deterioration, ranging from approximately 12.5% at the earliest mix identifiers to 8.8% at the later ones. Introducing 2% GNP by mass of cement reduced the sulphate attack values to about 11.2 to 7.9%, while 5% GNP by mass of cement mixes showed further improvement of values ranging between 10% and 6.7%. The 10% GNP by mass of cement mixes did even better, decreasing from 9% to 5.8%. The most favourable resistance was recorded for the 20% GNP by mass of cement mixes, which showed a significant reduction in damage, ranging from 8% to 5.1% and as low as 3.8% at the middle range of the mix identifier. Increasing the dosage of GNP results in a significant improvement in sulphate resistance, due to better densification and chemical stability of the matrix.

The enhancement of sulphate resistance due to GNP incorporation is consistent with earlier reported findings in nano-engineered cementitious systems. Graphene nanoplatelets significantly reduce pore connectivity and enhance the microstructural packing that limits the entry of sulphate ions and minimizes the chemical degradation⁵³. Several studies have shown that nanomaterials based on graphene improve the consistency of C-S-H and reduce the appearance of expansive products such as ettringite and gypsum products during sulphate exposure⁵⁴. The observed improvement in performance at higher dosages (10–20%) is consistent with reports that high nanoparticle content enhances barrier properties and reduces capillary porosity. Moreover, the superior performance of 20% GNP by mass in cement mixes is supported by the literature, which highlights that graphene derivatives promote early hydration, densify the ITZ, and form a more chemically stable microstructure resistant to ion diffusion. The slightly higher levels of sulphate damage with increasing mix ID across all dosages are consistent with trends observed in studies examining the effects of mix composition, w/c ratio, and dispersion

levels on long-term durability behaviour. The improvement in sulphate resistance and reduction in expansion are detailed in Fig. 12.

Rapid chloride penetration test (RCPT)

The RCPT values show a significant decrease in chloride-ion diffusion as GNP gratification increases for all mix IDs. The control mixes (0%) showed the highest charge passed, ranging from approximately 2800 coulombs for the earlier mix IDs to approximately 2100 coulombs for the later ones. The introduction of 2% GNP by mass of cement reduced the penetration to approximately 1800–2400 coulombs, whereas 5% GNP by mass of cement mixes showed further improvement with values between 1300 and 2000 coulombs. The 10% GNP by mass of cement mixes were even better, showing values in the range of 1600–950 coulombs. The best durability was noted in the 20% GNP by mass of cement mixes, which had the lowest penetration levels, ranging from 1200 to 750 coulombs, with as low as 650 coulombs at the mid-range mix IDs. These results suggest a significant increase in chloride resistance as GNP dosage increases.

The stable reduction in RCPT standards with increasing GNP content correlates with various studies in the literature, demonstrating that graphene nanoplatelets significantly reduce chloride ion permeability due to improved hole structure and increased matrix compactness⁵¹. GNPs act as nano-fillers that block capillary channels and reduce pore connectivity, thereby significantly limiting ion diffusion⁵¹. Their increased surface area also leads to faster hydration and C–S–H formation, thereby increasing resistivity and reducing ionic mobility. The lowest RCPT values measured at a 20% GNP by mass of cement dosage are in line with literature results, which show that higher nanomaterial content also improves impermeability as long as dispersion is adequately maintained⁵⁵. Additionally, the mid-series minimum for several mixes is associated with optimal hydration densification and gel formation, similar to the hydration- and microstructure-based durability mechanisms reported in recent research on nano-concrete. The notable reduction in chloride ion penetration with GNP incorporation is clearly proven in Fig. 13. The results indicate that the addition of Graphene Nanoplatelets resulted in a general improvement in the strength and durability of the different concrete grades. These results indicate that Graphene Nanoplatelets have increased the density of the cement matrix through pore refinement and increased the contact area at the interface between the paste and aggregate particles.

The improvement in durability performance observed in this study is consistent with the microstructural evidence obtained from SEM–EDS and XRD analyses (Sect. 4.11). The reduced pore connectivity and denser matrix observed in SEM images directly explain the lower ingress of water and aggressive ions. Additionally, the confirmed presence of well-developed hydration products supports the formation of a compact microstructure that restricts transport pathways. Therefore, the durability enhancements reported are not solely inferred from literature but are substantiated by the experimental microstructural characterization conducted in this study.

Experimental variability can arise from various sources, including the effectiveness of nanoparticle dispersion, aggregate packing behaviour, and curing conditions. Nonetheless, the results across multiple mix designs show consistent trends; therefore, it is reasonable to conclude that the documented performance enhancements resulted from the incorporation of graphene nanoplatelets into the matrix. These observed enhancements in both strength and durability are consistent with previous research demonstrating improvements in these properties for graphene-modified cement-based materials, further validating the experimental evidence presented in this paper. In addition to evaluating the macroscopic performance of graphene nanoplatelet-modified concrete, microstructural evaluation using techniques such as SEM and XRD could provide a deeper understanding of graphene nanoplatelet dispersion within the cement paste, the formation of hydration products, and changes in the pore size distribution of the hardened cement paste.

Microstructural observation (SEM–EDS and XRD analysis)

The mechanism by which the properties of mechanical performance and durability of the GNP-modified concrete are improved is investigated through microstructural examinations via Scanning Electron Microscopy (SEM) utilizing Energy Dispersive Spectroscopy (EDS) and X-ray Diffraction (XRD). A representative SEM image along with an EDS spectrum of the GNP-modified cementitious matrix are displayed in Fig. 14a and b, respectively, whereas the XRD patterns of the cement and GNP-modified concrete are provided in Fig. 14c. As indicated by the SEM images, the inclusion of GNP in the matrix produces a dense, compact microstructure with improved particle packing and reduced pore connectivity in the cementitious composite. Due to the nanofilling effect produced by the incorporation of graphene nanoplatelets, the microvoids and capillary pores in the matrix are occupied and capped by ultrathin layers of graphene nanoplatelets; thus, the pore structure of the matrix is refined, and its Permeability is reduced. Additionally, the layered morphology and high tensile strength of graphene nanoplatelets enable them to act as Crack Bridging Reinforcement and restrict the propagation of microcracks; therefore, the tensile and flexural performance of the composite material is enhanced. The EDS spectra associated with the SEM Images provide evidence of the presence of Major Elements such as Calcium (Ca), Silicon (Si), Oxygen (O), and Carbon (C) within the matrix. The presence of a majority of the Ca–Si–O indicates that the C–S–H Gel, which is the Primary Hydration Product responsible for the Strength Development in Cementitious Materials, has been formed. The presence of Carbon Peaks provides evidence of the inclusion of Graphene Nanoplatelets within the matrix. The XRD Patterns provide additional information about the diffraction peaks that correspond to Quartz, Calcium Silicate Phases and Other Crystalline Hydration Products Typically Observed in Hydrated Cement Systems. This provides evidence that the addition of Graphene Nanoplatelets does not alter the Fundamental Crystalline Phases but instead contributes to the Refining of the Microstructure, Improvement of the Densification of the Matrix, and Improvement of the Interfacial Bonding within the Cementitious Composite. Therefore, the microstructural observations support the experimental results from Mechanical and Durability Tests, demonstrating that the Combined Effects of Nano-Filling, Nucleation of

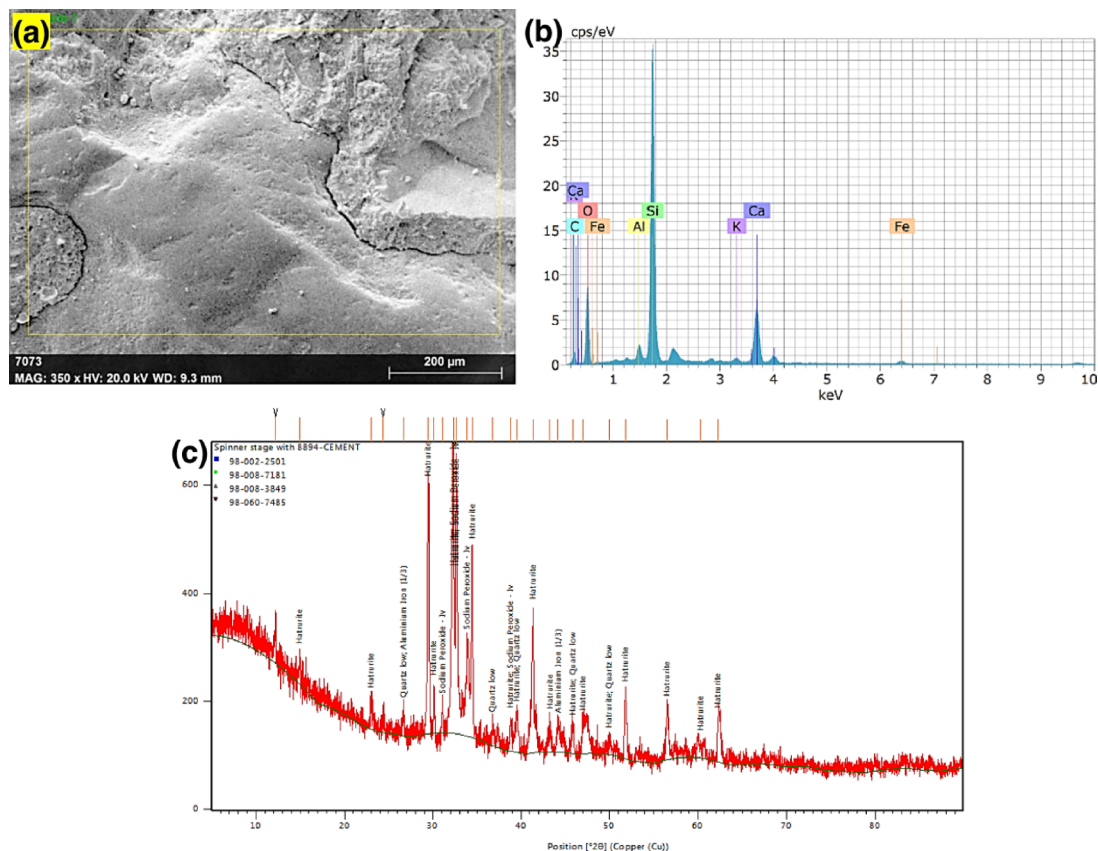


Fig. 14. Photographs of microstructural observations for (a) densification and morphology, (b) elemental composition and hydration confirmation, and (c) crystalline phase stability and structural refinement.

Hydration Products, Crack Bridging, and Improved Characteristics of the Interfacial Transition Zone (ITZ) contribute to the Enhanced Strength and Durability Performance of the GNP-modified concrete.

It is important to note that the microstructural observations presented in this study provide qualitative validation of the proposed mechanisms. While SEM-EDS and XRD analyses confirm matrix densification, hydration enhancement, and phase composition, more advanced quantitative techniques such as mercury intrusion porosimetry or transmission electron microscopy would be required for detailed characterization of pore size distribution and nanoscale crack-bridging behaviour. These aspects are recommended for future investigation to further strengthen the mechanistic understanding.

Conclusion

This study evaluated the influence of graphene nanoplatelets (GNPs) on the fresh, mechanical, and durability properties of concrete across multiple strength grades (M20–M40) and dosage levels (0–20%). The results demonstrate that increasing GNP content reduces workability due to higher surface area and water demand. Mechanical performance improved consistently with GNP incorporation, with notable increases in compressive, flexural, and split tensile strengths. These enhancements are attributed to nano-filling, accelerated hydration, and crack-bridging mechanisms. Durability characteristics also improved significantly, as evidenced by reduced water absorption, lower sulphate expansion, and decreased chloride-ion permeability, indicating a refined pore structure and enhanced matrix densification. Although maximum performance was observed at 20% GNP dosage, reduced workability at higher contents limits practical applicability. An optimal dosage range of 5–10% is recommended to balance performance and workability.

Data availability

The data supporting the findings of this study can be obtained from the corresponding author upon reasonable request.

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Author contributions

S. D. Anitha Selvasofia: Methodology, Investigation, Writing—original draft. Ayinala Naga Sai: Review and writing. A. Mohanraj: Review and writing. G. K. Arunvivek: Review and writing. Pramod Kumar: Investigation, Writing—original draft. Bheem Pratap: Review and writing. Regasa Yadeta Sembeta: Review and writing.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

This study did not involve human participants or animals; no ethical approval was required. All research procedures adhered to relevant ethical guidelines and best practices for non-human and non-animal research.

Declaration of AI use

An AI-based tool was used during the drafting process to enhance readability and language clarity. The authors subsequently reviewed and edited the content as necessary and take full responsibility for the final article.

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